

The Architecture of Hair

The strand you think you know — its layers, its cycle, and the life it lives beneath the skin

Learning Objectives

- Name and describe the three layers of a single hair strand and their functions
- Describe the structure of the follicle beneath the surface, and the glands that serve it
- Explain the four phases of the hair growth cycle and roughly how long each lasts
- Explain why keratin's chemical bonds are the foundation of every service you will ever perform on hair

Before technique, understanding

Every hand that will ever touch a client's hair at Amber must first understand, completely, what it is touching. Not a costume. Not an accessory that happens to grow from the scalp, though the world often treats it that way. A living structure — dead in its visible length, yes, but the product of a follicle that is very much alive, and a process that has been running quietly beneath the skin since before the client was born. This module is the foundation everything else in the Atelier's hair curriculum is built upon.

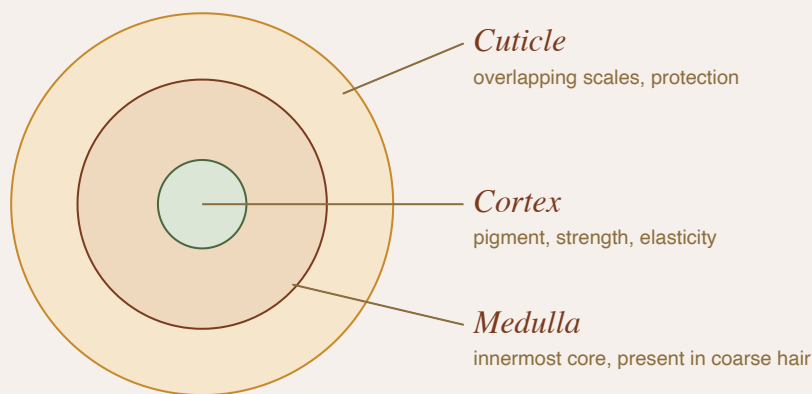
You cannot honour what you do not understand. Understanding is the first form of respect a practitioner owes the hair.

One strand, three layers

What you see and hold in your hands is the hair shaft — the visible, non-living portion of the hair. Cut through it, and you would find three concentric layers, much like the rings of a tree.

- **The cuticle** — the outermost layer, made of overlapping, translucent scales, much like tiles on a roof. Its job is purely protective: it shields the layers beneath, and its condition determines whether hair looks shiny or dull, feels smooth or rough.
- **The cortex** — beneath the cuticle, and the bulk of the strand's mass. It contains the pigment melanin, which gives hair its colour, and the majority of the structural proteins that give hair its strength and elasticity. Every chemical service — colour, perm, relaxer — works primarily in this layer.
- **The medulla** — the innermost core, present in thicker hair and often absent in fine hair. Its function is still not fully understood, though it is believed to play a minor role in heat regulation.

◆ Figure 1.1 · The Three Layers of a Strand



A cross-section of a single hair strand, viewed from the tip — cuticle, cortex, and medulla, ring within ring.

Beneath the surface: the follicle

The visible shaft is only the story's ending. Its beginning is the follicle, a tunnel-shaped structure rooted in the dermis. At its base sits the hair bulb, wrapped around the dermal papilla — a small cluster of blood vessels that feeds the growing hair with oxygen and nutrients. It is the papilla, not the visible scalp, that truly determines the health of what eventually reaches your hands. Attached to each follicle is a sebaceous gland, producing sebum to condition the strand and scalp, and a tiny arrector pili muscle, responsible for the strand standing on end when the skin is cold or startled.

The four phases of growth

Hair does not grow continuously and uniformly. Each follicle moves through a cycle, independently of its neighbours — which is precisely why you never go bald overnight.

- **Anagen (growth)** — the active phase, lasting anywhere from two to seven years depending on genetics. At any given time, roughly 85–90% of scalp hair is in anagen.
- **Catagen (transition)** — a brief two-to-three-week phase in which the follicle shrinks and detaches from its blood supply. Only about 1% of hair is in catagen at any time.
- **Telogen (rest)** — a resting phase lasting roughly three months, during which the strand is fully formed but no longer growing. Around 10–15% of hair is in telogen at any time.
- **Exogen (shedding)** — the shedding of the resting strand, making way for a new anagen hair to begin the cycle again. Losing 50–100 strands a day through exogen is entirely normal — you will return to this number, and to when it becomes a warning sign, in Module Eight.

The chemistry beneath the craft

Hair is roughly 91% protein — almost entirely keratin, the same fibrous protein family that forms nails and the outermost layer of skin. Keratin's strength comes from three types of chemical bonds holding its structure together: hydrogen bonds and salt bonds, which are weak and temporarily broken by water or heat — the entire basis of blow-drying and thermal styling — and disulfide bonds, which are strong, permanent, and can only be broken and reformed by chemical means. Every relaxer, perm, and chemical straightening service you will study in Module Four works by breaking and rebuilding disulfide bonds. Understanding this one fact is what separates a practitioner who follows a formula from one who understands why the formula works.

The scalp is skin

It is easy to think of the scalp as mere real estate — the ground hair happens to grow from. It is not. The scalp is skin, with all the same layers, glands, and living complexity you would study in a facial curriculum, plus a density of follicles found nowhere else on the body. It carries its own scalp microbiome — a living community of bacteria and fungi that, when balanced, helps regulate oil production and defend against conditions like dandruff and folliculitis. The old model of haircare treated the scalp as something to be stripped and controlled with harsh

cleansers. The Amber model treats it as a living ecosystem to be tended — the same philosophy you were taught for skin in Course I, now extended to the crown.

Key Terms

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[object Object]

1

[object Object]

2

[object Object]

3

[object Object]

4

[object Object]

Practical Activity · Draw the Crown

Sketch a labelled cross-section of a hair strand and its follicle — all three layers, the bulb and papilla, and the sebaceous gland. Beside your drawing, write one sentence per structure in your own words. **Keep this drawing. It is the first page of your capstone portfolio in Module Fifteen.**